

Contents

Count Lines in a File	2
Copy File Content.....	4
Count Words in a File.....	6
Find Longest Line.....	8
Count Specific Word Occurrence	10
Replace Word in File	12
Count Character Frequency	14
Append Text to a File	16
Read File in Reverse.....	18
Find and Count Word Occurrence in File	20

Count Lines in a File

Scenario:

You are given a text file, and your task is to count the number of lines in the file.

Problem Statement:

Write a Python function that reads a file and returns the total number of lines in it.

Constraints:

- The file may have up to 10^5 lines.

Test Cases 1:

Input File Content:

```
Hello, World!
Welcome to Python programming.
This is a simple file.
```

Output: 3

Test Cases 2:

Input File Content:

```
Python is fun.
Learning file handling.
```

Output: 2

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output: 0
- **Input File Content:** "One line only."
Output: 1
- **Input File Content:**

```
Line 1
Line 2
Line 3
Line 4
Line 5
```

Output: 5

- **Input File Content:**

Line 1
Line 2
Line 3
Line 4

Output: 4

- **Input File Content:**

Line 1
Line 2
Line 3

Output: 3

Copy File Content

Scenario:

You are given a text file, and your task is to copy its content to another file.

Problem Statement:

Write a Python function that reads a file and writes its content to another file.

Constraints:

- The input file may contain up to 10^6 characters.

Test Cases

Input File Content:

```
Copy this text to another file.
```

Output File Content:

```
Copy this text to another file.
```

Test Cases

Input File Content:

```
This is a test file.  
It contains multiple lines.
```

Output File Content:

```
This is a test file.  
It contains multiple lines.
```

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output File Content: ""
- **Input File Content:** "Single line of text."
Output File Content: "Single line of text."
- **Input File Content:**

```
Line A  
Line B
```

Output File Content:

```
Line A
```

Line B

- **Input File Content:**

Line A
Line B
Line C
Line A
Line B
Line C

Output File Content:

Line A
Line B
Line C
Line A
Line B
Line C

- **Input File Content:**

Line A
Line B
Line C

Output File Content:

Line A
Line B
Line C

Count Words in a File

Scenario:

You need to determine the total number of words in a given text file.

Problem Statement:

Write a Python function that reads a file and returns the total number of words.

Constraints:

- The file may contain up to 10^6 words.

Test Cases

Input File Content:

```
Count the number of words in this file.
```

Output: 8

Test Cases

Input File Content:

```
This file has five words total.
```

Output: 6

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output: 0
- **Input File Content:** "Just one word."
Output: 3
- **Input File Content:**

```
Python file handling is useful.
```

Output: 5

- **Input File Content:**

```
GLA University.
```

Output: 2

- **Input File Content:**

This is the file content.

Output: 5

Find Longest Line

Scenario:

You need to find the longest line in a text file based on the number of characters.

Problem Statement:

Write a Python function that reads a file and returns the longest line.

Constraints:

- The file may contain lines with up to 10^3 characters each.

Test Cases

Input File Content:

```
Short line.  
This is a much longer line in the file.  
Another line.
```

Output: "This is a much longer line in the file."

Test Cases

Input File Content:

```
Python is fun.  
Handling files is interesting.
```

Output: "Handling files is interesting."

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output: ""
- **Input File Content:** "Single line."
Output: "Single line."
- **Input File Content:**

```
Line with some text.  
Another line with more text.  
A much longer line than the previous ones.
```

Output: "A much longer line than the previous ones."

- **Input File Content:**

Line with some text.
Another line with more text.

Output: "Another line with more text."

- **Input File Content:**

Line with some text.
Another line with more text.
A much longer line than the previous ones.
An another much longer line than the previous ones.

Output: "An another much longer line than the previous ones."

Count Specific Word Occurrence

Scenario:

You need to count how many times a specific word appears in a text file.

Problem Statement:

Write a Python function that reads a file and counts the occurrences of a specific word, irrespective of the case (uppercase or lowercase).

Constraints:

- The word search is case-sensitive.
- The file may contain up to 10^5 words.

Test Cases

Input File Content:

```
The quick brown fox jumps over the lazy dog.  
The fox is quick.
```

Word to Count: fox

Output: 2

Test Cases

Input File Content:

```
This is a simple test file.  
Test the file with various words.
```

Word to Count: test

Output: 2

Additional Test Cases:

- **Input File Content:** "" (empty file), **Word to Count:** any
Output: 0
- **Input File Content:** "A word in a single line.", **Word to Count:** word
Output: 1
- **Input File Content:**

```
Python is great for file handling.  
File handling in Python is easy.
```

Word to Count: Python

Output: 2

- **Input File Content:**

```
Python is great for file handling.  
File handling in Python is easy.
```

Word to Count: is

Output: 1

- **Input File Content:**

```
Python is great for file handling.  
File handling in Python is easy.  
File handling is easy to understand.
```

Word to Count: File

Output: 3

Replace Word in File

Scenario:

You have a text file and need to replace all occurrences of a specific word with another word.

Problem Statement:

Write a Python function that reads a file, replaces all occurrences of a given word with another word, and writes the updated content to a new file.

Constraints:

- The file may contain up to 10^5 words.
- The word replacement should be case-sensitive.

Test Cases

1. Input File Content:

```
The cat sat on the mat.  
The cat is cute.
```

Word to Replace: cat, **Replacement Word:** dog

Output File Content:

```
The dog sat on the mat.  
The dog is cute.
```

2. Input File Content:

```
Python is fun.  
Python is powerful.
```

Word to Replace: Python, **Replacement Word:** Java

Output File Content:

```
Java is fun.  
Java is powerful.
```

Additional Test Cases:

- **Input File Content:** "" (empty file), **Word to Replace:** anything, **Replacement Word:** something
Output File Content: ""
- **Input File Content:** "Hello world.", **Word to Replace:** world, **Replacement Word:** universe
Output File Content: "Hello universe."
- **Input File Content:**

A quick brown fox jumps over the lazy dog.
The fox is quick.

Word to Replace: fox, **Replacement Word:** wolf
Output File Content:

A quick brown wolf jumps over the lazy dog.
The wolf is quick.

- **Input File Content:**

A quick brown fox jumps over the lazy dog.
The fox is quick.

Word to Replace: quick, **Replacement Word:** slow
Output File Content:

A slow brown fox jumps over the lazy dog.
The fox is slow.

- **Input File Content:**

A quick brown fox jumps over the lazy dog.
The fox is quick.

Word to Replace: dog, **Replacement Word:** cat
Output File Content:

A quick brown fox jumps over the lazy cat.
The fox is quick.

Count Character Frequency

Scenario:

You need to analyze the frequency of each character in a text file.

Problem Statement:

Write a Python function that reads a file and returns a dictionary with the frequency of each character.

Constraints:

- The file may contain up to 10^6 characters.
- The analysis should be case-sensitive.

Test Cases

1. Input File Content:

```
Hello, World!
```

Output:

```
{'H': 1, 'e': 1, 'l': 3, 'o': 2, ',': 1, ' ': 1, 'W': 1, 'r': 1, 'd': 1, '!': 1}
```

2. Input File Content:

```
Python programming.
```

Output:

```
{'P': 1, 'y': 1, 't': 1, 'h': 1, 'o': 2, 'n': 2, ' ': 1, 'p': 1, 'r': 2, 'g': 2, 'a': 1, 'm': 2, '.': 1}
```

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output: {}
- **Input File Content:** "A"
Output: {'A': 1}
- **Input File Content:** "aaa bbb ccc"
Output: {'a': 3, ' ': 2, 'b': 3, 'c': 3}
- **Input File Content:** "aaa bbb ccc ddd"
Output: {'a': 3, ' ': 2, 'b': 3, 'c': 3, 'd': 3}
- **Input File Content:** "aaa bbb cc"
Output: {'a': 3, ' ': 2, 'b': 3, 'c': 2}

Append Text to a File

Scenario:

You need to append additional text to an existing file.

Problem Statement:

Write a Python function that appends a given string to the end of a file.

Constraints:

- The file may initially contain up to 10^6 characters.

Test Cases

1. **Initial File Content:** Original text. **Text to Append:** More text.
Output File Content: Original text. More text.
2. **Initial File Content:**

```
Line 1
Line 2
```

Text to Append: Appended line.

Output File Content:

```
Line 1
Line 2
Appended line.
```

Additional Test Cases:

- **Initial File Content:** "" (empty file), **Text to Append:** New content.
Output File Content: New content.
- **Initial File Content:** "Start.", **Text to Append:** " End."
Output File Content: "Start. End."
- **Initial File Content:**
Before append.
Text to Append:
After append.
Output File Content:
Before append.
After append.
- **Initial File Content:**
Before append.
Line 1
Text to Append:
After append.
Line 2
Output File Content:

Before append.

Line 1

After append.

Line 2

- **Initial File Content:**

Before append.

Line 1

Text to Append:

After append.

Output File Content:

Before append.

Line 1

After append.

Read File in Reverse

Scenario:

You need to read a file and return its content in reverse order, line by line.

Problem Statement:

Write a Python function that reads a file and returns its content in reverse order.

Constraints:

- The file may contain up to 10^5 lines.

Test Cases

1. Input File Content:

```
Line 1
Line 2
Line 3
```

Output:

```
Line 3
Line 2
Line 1
```

2. Input File Content:

```
First line.
Second line.
Third line.
```

Output:

```
Third line.
Second line.
First line.
```

Additional Test Cases:

- **Input File Content:** "" (empty file)
Output: ""
- **Input File Content:** "Single line."
Output: "Single line."
- **Input File Content:**

```
Apple
Banana
Cherry
```

Output:

Cherry
Banana
Apple

- **Input File Content:**

A
B
C

Output:

C
B
A

- **Input File Content:**

Apple
Box
Choice
Battle

Output:

Battle
Choice
Box
Apple

Find and Count Word Occurrence in File

Scenario:

You need to find a specific word in a file and count how many times it occurs.

Problem Statement:

Write a Python function that reads a file and counts how many times a specific word appears.

Constraints:

- The word search should be case-sensitive.
- The file may contain up to 10^6 words.

Test Cases

Input File Content:

```
The cat sat on the mat.  
The cat is cute.
```

Word to Count: cat

Output: 2

Test Cases

Input File Content:

```
Python is fun.  
Python is powerful.
```

Word to Count: Python

Output: 2

Additional Test Cases:

- **Input File Content:** "" (empty file), **Word to Count:** anything
Output: 0
- **Input File Content:** "Hello world.", **Word to Count:** world
Output: 1
- **Input File Content:**

```
Quick brown fox jumps over the lazy dog.  
The fox is clever.
```

Word to Count: fox

Output: 2

- **Input File Content:**

Quick brown fox jumps over the lazy dog.
Quick brown fox jumps over the lazy dog.
Quick brown fox jumps over the lazy dog.

Word to Count: fox

Output: 3

- **Input File Content:**

This is the file content line number one.
This is the file content line number two.
This is the file content line number three.

Word to Count: file

Output: 3